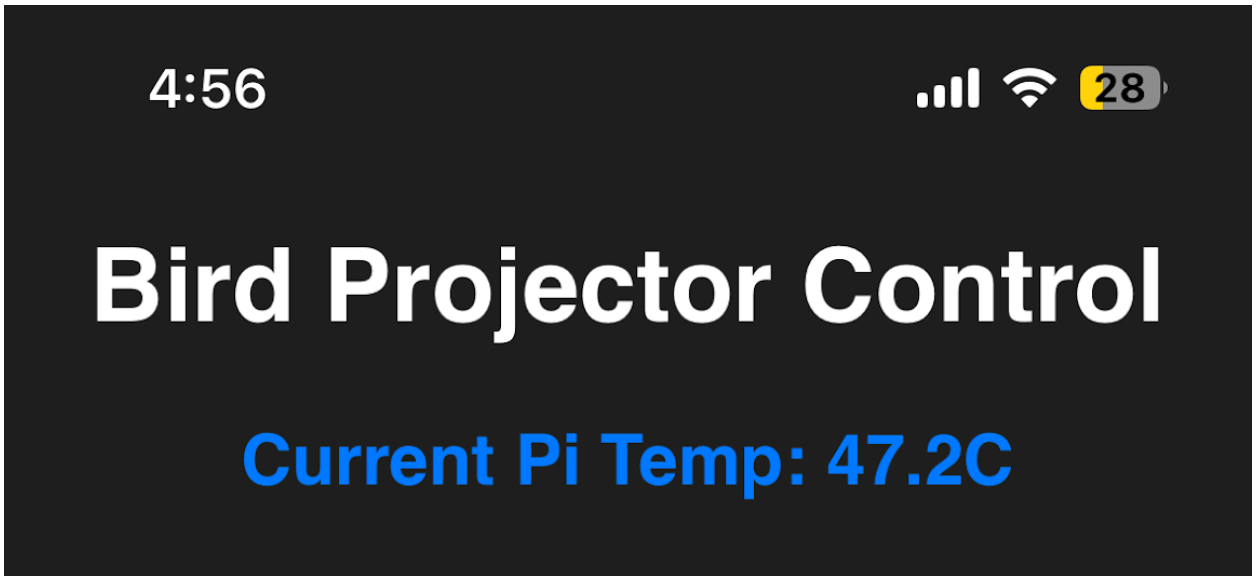
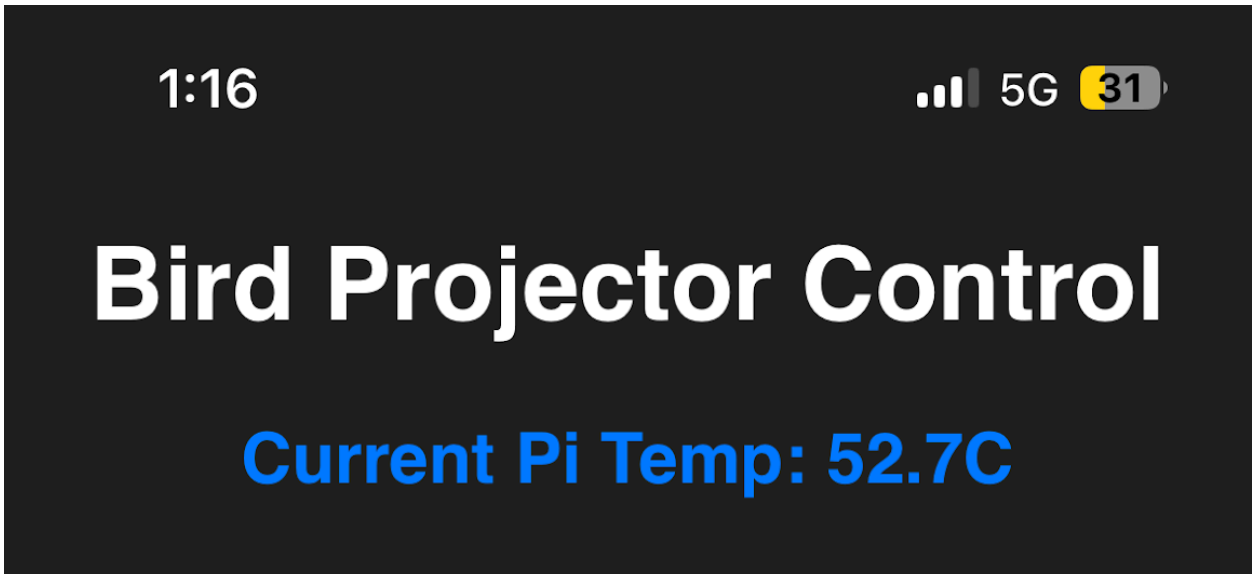
Heat Control Testing:
This is the test for the pi just running normally for an extended period of time

This shows us first starting the pi and just having it running normally



This is after about 3 hrs it shows that the temp stays at a good temp and only slightly climbs and then levels out

After running it overnight for an overall time of about 20 hrs we kept a good temp and hover around 55-60 degrees celsius depending on the room / usage



The goal is to ensure the bird-system.service reliably starts and plays the primary deterrent video without manual intervention upon receiving power. A cold boot was initiated by connecting the main extension cord to a power source. The time to the first frame of the video display was recorded.

Test Data:

Trial	**Action**	**Time to Play**
1	Connect Power	22.4 seconds
2	Connect Power	24.1 seconds
3	Connect Power	21.8 seconds
4	Connect Power	23.5 seconds
5	Connect Power	22.9 seconds

Ensure the housing can withstand harsh environmental interactions, such as high winds or animals landing on/striking the case. Simulated environmental stress through blunt force and weight loading.

Test Data:

Test Type	**Simulation**	**Result**
Kinetic Shock	Excessive shaking / blunt impacts (simulating animals/wind)	No internal dislodgment, casing remained intact.
Static Load	10 lbs placed on top (simulating resting animals)	Zero structural damage.
Static Load	25 lbs placed on top (Reference Figure 8 from report)	Zero structural damage, seals intact.

Validate that the system can operate safely outdoors during rain and harsh weather. Direct water splashing and spraying were applied to the case seams and sealed points.

Test Data:

Test Area	**Application**	**Observation**
Top/Side Seams	Moderate water splashing	Water beaded and rolled off.
Sealed Points	Direct spray (Simulating heavy rain/wind)	No water ingress detected internally.
Internal Electronics	Post-spray inspection	Completely dry, 100% operational.